Analysis of the SPARC Galactic Catalog With Modified Gravity

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(Dated: August 13, 2026)

I. GALACTIC-ROTATION CURVES

In this paper, Eqs. (1) and (2) below are applied to a selection of nearby galaxies from the SPARC (Spitzer Photometry & Accurate Rotation Curves) database to derive their rotation curves, which are then checked against observations:

$$H_f = \sqrt{\frac{v_f}{c}} H_0 \ , \tag{1}$$

$$a_n = \frac{m_r G}{r} \left(\frac{1}{r} + \frac{2\sqrt{cH_f}}{\sqrt{m_t G}} \right) \ . . \tag{2}$$

The SPARC database comprises a large catalog of galaxies that vary in mass, gas fraction, effective radii, and surface brightness [1]. As a cohesive dataset spanning a broad range of morphologies, the SPARC data provide an ideal test of modified-gravity as well as dark-matter halo models, the latter being the primary focus of the cited paper. The complete SPARC catalog contains too many galaxies to analyze here—where the intent is a quick sanity check of the above equations. A subset of the catalog was selected to be examined based on proximity to the Milky Way—with the cutoff set to 5 Mpc. These galaxies are listed in Tables I and II of Appendix III. This appendix also includes the corresponding rotation curve for each galaxy as generated via Eq. (2). Distance estimates, listed in Table I, were sourced from the *NASA/IPAC Extragalactic Database* [2].

Appendix II is a synopsis of the procedure used to generate the rotation curves of Appendix III. In short, discs are modeled as zero-thickness plates of varying density with respect to radius. Bulges (one galaxy in the selection has a bulge—NGC 6946) are modeled as spheres where density varies according to radius as well. Discs and bulges are broken into finite elements, and Eq. (2) is solved for each element, where r is the distance from the center of mass of that element to an offset from the center of rotation. The net acceleration a_n—toward the center of rotation—at a given offset is the vector sum of the accelerations with respect to each disk and bulge element. Net acceleration a_n is solved across a range of offsets for a given galaxy from the inner to the outermost data points—thereby establishing a rotation curve. The outermost data point, for each galaxy, is listed under the "r_{max}" column in Table I.

Looking at Eq. (2), when applied to galaxies, there are two free parameters: m_t, which is the total galactic mass, and H_f, the expansion parameter of the moving frame containing the galaxy. The total mass m_t is constrained by estimates of the matter comprising the stellar populations, dust, and gas. In practice, H_f is adjusted to fit a given galaxy's rotation curve, but the resulting value should comport with Eq. (1). This equation has two free parameters: v_f and H_0, which are only free to a certain extent. Even considering the Hubble tension, estimates of H_0 fall within a relatively narrow range. The same holds for the peculiar speed v_f of a body, which can be gauged via the CMB dipole and relative motions.

Each galaxy in the SPARC database includes curve fits for 12 different dark-matter halo models. To fit a given model, SPARC researchers adjusted the following free parameters: bulge mass-to-light ratio (Υ_b), stellar-disk mass-to-light ratio (Υ_d), galaxy distance, and disk inclination. This highlights another advantage of the SPARC catalog—as fitting the sundry dark-matter models required adjustments to these parameters, yielding a variety of stellar-mass models per galaxy from which to source data.

Looking at the two rotation curves for NGC 2976 in Appendix III, for example, the curve with an $\Upsilon_d = 0.48$ was generated via the Burkert-Flat model (referred to simply as the "Burkert model" going forward), whereas the curve with an $\Upsilon_d = 0.60$ was generated via the DC14-LCDM model. The stellar-disk Newtonian curves in these figures—drawn as dashed blue lines—come directly from the SPARC catalog. The curves show the Keplerian decline that would be expected with only the stellar-disk mass as the source of centripetal acceleration—where that acceleration is purely Newtonian gravity. Notice that the stellar-disk curve from the Burkert model, with a smaller Υ_d, looks like a scaled-down version—vertically—of the DC14-LCDM model's curve.

In addition to the stellar disk, Newtonian curves for the gas disk and the bulge (where appropriate) are published in the SPARC catalog for each dark-matter model of each galaxy. Table I of Appendix III relates a given galaxy to the dark-matter model used as the source of Newtonian curves for the components of that galaxy. These curves were translated to density profiles and fed into the finite-element analysis of Appendix II—generating the net rotation curve for the galaxy in question. All source Newtonian curves from the SPARC catalog were reproduced in the figures of Appendix III. The Newtonian contribution from the gas disk is drawn as a dashed magenta line (with the larger dashes), and the bulge contribution is shown as a dotted green line. The net rotation curve is shown as a solid cyan line.

Variance in the free parameters of distance and inclination can result in the error bars of one model looking offset or scaled (or both) relative to those of another model with respect to the axes of distance and speed. This can be seen when comparing the Burkert and DC14-LCDM models of NGC 2976—where the error bars of the Burkert model peak at roughly 88 km s^{-1} at 2.1 kpc vs. a maximum speed of 81 km s^{-1} at 2.2 kpc for the error bars of the DC14-LCDM model.

The gas curves of these two models both peak at roughly 30 km s^{-1} but at different distances—with the Burkert curve peaking at 1.75 kpc vs. 1.8 kpc for the DC14-LCDM curve. The gas curves for a given galaxy are more consistent between models than the stellar curves, which are subject to being modified via the additional free parameter of mass-to-light ratio.

The rotation curves of the galaxies in Appendix III were generated via the dark-matter model offering the best fit—as referenced in Table I. With density profiles for the stellar and gas disks (and bulge when relevant)—derived from the given Newtonian curves for these components, the expansion parameter H_f for a given galaxy is calibrated to fit the net rotation curve to the error bars associated with the source model. Referring, again, to NGC 2976, notice that the Burkert model (with a lower $\Upsilon_d = 0.48$) prefers the higher H_f of 1.9 than the DC14-LCDM model (with an $\Upsilon_d = 0.60$)—where fitting the curve favors an H_f of .20. It's concluded that, for NGC 2976, the expansion parameter H_f lies somewhere between .20 and 1.9 km s^{-1} Mpc^{-1}.

Table II shows the expansion parameter H_f for each galaxy in Table I used to generate that galaxy's rotation curve. Table II also includes the galactic mass m_t as well as the frame constant k_f—defined by Eq. (5)—for each galaxy, which is computed via H_f and m_t. The galactic mass m_t is calculated via the density curves of the associated dark-matter model. Because m_t is derived from data gleaned from the SPARC catalog, m_t is only a free parameter to the extent that there are 12 models per galaxy from which to source density curves. Out of the 29 galaxies listed in Table I, 12 used density profiles derived from the Burkert model, 5 used density profiles derived from the coreNFW LCDM model, and the rest used a combination of the remaining models.

Solving Eq. (1) for v_f for the cases where $H_f = .20$ and 1.9 gives the following:

$$v_f = c\left(\frac{H_f}{H_0}\right)^2 ,$$

$$v_f(.2) = 299\,792\,\mathrm{km\,s^{-1}}\left(\frac{.2}{70}\right)^2 = 0.3\,\mathrm{km\,s^{-1}} ,$$

$$v_f(1.9) = 299\,792\,\mathrm{km\,s^{-1}}\left(\frac{1.9}{70}\right)^2 = 221\,\mathrm{km\,s^{-1}} .$$

This predicts a peculiar speed v_f for NGC 2976 between .3 and 221 km s^{-1}.

As touched upon earlier, an observer with a peculiar speed will see a dipole in the CMB. In the case of the Milky Way, the amplitude of this dipole suggests that our galaxy has a speed of 565 km s^{-1} relative to the CMB [3].

Similarly, if the motion of a galaxy relative to the Milky Way is known, the peculiar speed of that galaxy relative to the CMB can be deduced. However, while the recessional component of a galaxy's relative velocity can be resolved by measuring its redshift—while accounting for cosmic redshift, determining the transverse component of the galaxy's velocity is more challenging [4].

If, for example, a galaxy were found along the axis of the Milky Way's dipole moving away from the blueshifted side of the dipole with a peculiar speed of 565 km s^{-1}, it could be deduced that the galaxy was at rest relative to the CMB frame—but only in the radial direction. If the galaxy had a transverse velocity, it would have a dipole perpendicular to that of the Milky Way.

Conversely, if a galaxy were found perpendicular to the axis of the Milky Way's dipole with zero recessional peculiar speed, it could be concluded to be moving in parallel with the Milky Way—with a peculiar speed of 565 km s^{-1}—but only if its transverse velocity were verified to be zero.

It's surmised, however, that the speeds of galaxies with respect to the CMB that are derived purely from redshift data—without resolution of those galaxies' transverse velocities relative to us—are still useful data. Given a large enough data set, if Eq. (1) is valid, it seems reasonable that data points should concentrate along a trend line that comports with this relation.

Peculiar speeds relative to the CMB—based on redshift data—of the galaxies in Appendix III are listed in Table I under the column, "CMB redshift". These speeds were sourced from the *NASA/IPAC Extragalactic Database*. Notice that the CMB redshift for NGC 2976 is 90 km s^{-1}, which is nearly in the middle of the above estimated range: 0.3–221 km s^{-1}.

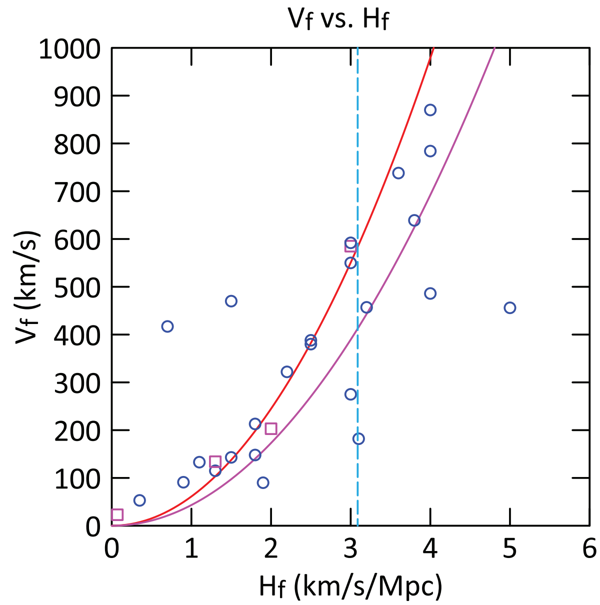


FIG. 1.

Fig. 1 is a plot of peculiar speed v_f vs. expansion parameter H_f for the galaxies in Appendix III. The data points come from Tables I and II—with the value for v_f approximated by the CMB-redshift estimate published for a given galaxy and where the value for H_f is that which gives the best fit for that galaxy's rotation curve. Circles in the figure indicate fair to excellent curve fits, while squares denote poor curve fits. Four galaxies have poor fits, which are highlighted in Table II. This table also notes the outlying galaxies relative to the data trend in the plot.

The left curve in the figure, drawn in red, traces Eq. (1). Galaxies plotted in Fig. 1 proximate to the left curve—assuming accurate values for H_f—have CMB redshifts that are nearly equal to their true peculiar velocities. A second curve to the right, drawn in magenta, accounts for uncertainty in the data—including that which stems from transverse velocities being unresolved. It's reckoned that the latter would tend to leave peculiar velocities underestimated. The right curve relates the true peculiar speed v_f to H_f as follows—where v_2 is the published CMB redshift:

$$
\begin{aligned}
{v_f}^2 &= {v_2}^2 + {v_2}^2 \ , \\
v_f &= \frac{1}{\sqrt{2}} \ .
\end{aligned}
$$

A galaxy plotted near the right curve thus has a true peculiar speed v_f that is approximately $\sqrt{2}$ times greater than the published CMB redshift. This scalar is the ratio of the magnitude of a vector—comprised of two orthogonal components of equal length—to the length of either component. This scalar seems to be congruent with variation in the present dataset—as about 83% of the galaxies plotted in Fig. 1 are near or contained within the left and right curves. In general, the data trend seems to comport with Eq. (1).

Looking at the rotation curves in Appendix III, a similar percentage (about 86%) of the galaxies listed in Table II have good fits. The quality of fit for these galaxies was determined visually—no statistical analysis was performed. Again, the intent here was to present a thesis with basic sanity checks made along the way.

NGC 2915 is an example of a galaxy having both a poor fit and, further, one that is significantly off trend in Fig. 1. With an H_f of 12, the (H_f, v_f) coordinate for NGC 2915 is outside the bounds of the figure. It's not clear if this galaxy has a high peculiar speed, which would translate to a high H_f, or has more mass than estimated in the SPARC data, or both.

Poor curve fits don't necessarily indicate a discrepancy in galactic mass, as the distribution thereof could also be at issue. Referring again to NGC 2976, the stellar disks in the Burkert and DC14-LCDM models ($\Upsilon_d = .48$ and $\Upsilon_d = .60$, respectively) have different masses, but both yield rotation curves with good fits. This example also highlights the sensitivity that H_f has with respect to mass—where just a 25% increase in Υ_d translates to a nearly 90% reduction in H_f.

The rotation curves of two galaxies were fit using disk mass-to-light ratios that were not specifically cited in one of the 12 dark-matter models. For NGC 0247, an Υ_d of 1.0 was used, which is about midway between the DC14 flat and coreNFW LCDM models: 0.50 and 1.52, respectively. The error bars for NGC 0247 came from the coreNFW LCDM model.

For NGC 6789, an Υ_d of 1.0 was used as well, which is about the midpoint of that of the Lucky13 LCDM and DC14 LCDM models: 0.69 and 1.20, respectively. The error bars for NGC 6789 were sourced from the DC14 LCDM model.

As r goes to infinity in Eq. (2), the reference mass m_r approaches the total mass m_t—as the galaxy looks increasingly like a point mass to an orbiting particle. Thus, at large distances, Eq. (2) reduces to Eq. (3) as follows:

$$
a_n = \frac{m_r G}{r}\left(\frac{1}{r} + \frac{2\sqrt{cH_f}}{\sqrt{m_t G}}\right) \ , \tag{3}
$$

$$
\begin{aligned}
& m_r \to m_t \ , \\
& a_n = \cancelto{0}{\frac{m_t G}{r^2}} + \frac{2\sqrt{cH_f m_t G}}{r} \ , \\
& a_n = \frac{2\sqrt{cH_f m_t G}}{r} \ , \\
& \frac{{v_{flat}}^2}{\cancel{r}} = \frac{2\sqrt{cH_f m_t G}}{\cancel{r}} \ ,
\end{aligned}
$$

$$
v_{flat} = \sqrt{2}(cH_f m_t G)^{\frac{1}{4}} \ . \tag{4}
$$

Note that as m_t approaches a point mass, the local expansion parameter H_l—as seen by a particle of negligible mass in the field—approaches the frame's expansion parameter H_f. Further, in this regime, the Newtonian component of Eq. (3) loses significance in relation to the expansion component and thus, at a certain distance, this can be ignored. Here, Eq. (3) reduces to Eq. (10).

Assuming circular motion, Eq. (4) gives the asymptotic speed v_{flat} that is approached by particles as the expansion component of centripetal acceleration becomes dominant. Table II includes v_{flat} calculations for each galaxy via this equation.

1. *The Milky Way*

Fitting the Milky Way's rotation curve is left as a future exercise, but with an estimate of its peculiar speed v_f and (baryonic) mass m_t [6], the following parameters are easily calculated: H_f, k_f, and v_{flat}.

$$
\begin{aligned}
m_t &= 8.43 \times 10^{10}\,\mathrm{M}_\odot = 1.68 \times 10^{41}\,\mathrm{kg} \ , \\
v_f &= 565\,\mathrm{km\,s^{-1}} \ , \\
H_f &= 3.04\,\mathrm{km\,s^{-1}Mpc^{-1}} \ , \\
k_f &= 5.88 \times 10^{-61}\,\mathrm{kg^{-1}s^{-1}} \ , \\
v_{flat} &= 191\,\mathrm{km\,s^{-1}} \ .
\end{aligned}
$$

The parameter v_{flat} predicts that the Milky Way's rotation curve flattens to 191 km s^{-1}, which comports with recent observations that extend out to 25 kpc [7][8]. Yoshiaki Sofue has compiled a "grand" rotation curve for the Milky Way that extends out to 400 kpc [9]. From 40 to 100 kpc, v_{flat} aligns well with this curve. Beyond 100 kpc, uncertainty in the data increases, and the error bars encompass a wide range of rotational velocities—from roughly 100 to 200 km s^{-1}.

The frame constant k_f, defined as the expansion parameter per unit mass, applies to all bodies in the system, which, in the case of the Milky Way, would include the Solar System. With the definition of k_f given by Eq. (5), Eq. (2) can be simplified as follows:

$$k_f = \frac{H_i}{m_i} = \frac{H_f}{m_t} \ , \tag{5}$$

$$a_n = \frac{m_r G}{r}\left(\frac{1}{r} + 2\sqrt{\frac{c}{G}k_f}\right) \ , \tag{6}$$

$$a_n = \frac{m_r G}{r^2} + \frac{2}{r}\sqrt{\frac{c}{G}k_f} \ , \tag{7}$$

$$k_e = \frac{2}{\cancel{r}}\sqrt{\frac{c}{G}k_f}\left(\frac{r^{\cancel{2}}}{m_r G}\right) \ , \tag{8}$$

$$k_e = \frac{2r}{m_r G}\sqrt{\frac{c}{G}k_f} \ . \tag{9}$$

Plugging Eq. (5) into Eq. (2) gives Eq. (7), which defines a_n as the sum of the two components of acceleration—the Newtonian and the expansion components. k_e, in Eq. (9), is the ratio of these components, which is an indication of their relative strength. Notice that k_e, which is a function of r, shows that the transition from the Newtonian to the non-Newtonian regime is linear with respect to distance.

When applied to the Solar System, Eq. (9) shows that the acceleration regime is entirely Newtonian:

$$\begin{aligned} &m_r = 1\,\mathrm{M_\odot} = 1.988\,47 \times 10^{30}\,\mathrm{kg} \ , \\ &r = 1\,\mathrm{AU} = 149\,597\,870.7\,\mathrm{km} \ , \\ &k_e = 3.66 \times 10^{-30} \ . \end{aligned}$$

At the average radius of Earth's orbit, the Newtonian component is 30 orders of magnitude larger than the non-Newtonian component.

At 4.25 ly—the distance to Proxima Centauri, the expansion component is still negligible at 25 orders of magnitude less than the Newtonian component:

$$\begin{aligned} &r = 4.2465\,\mathrm{ly} = 4.017 \times 10^{13}\,\mathrm{km} \ , \\ &k_e = 9.84 \times 10^{-25} \ . \end{aligned}$$

Clearly, the effects of modified gravity are not detectable in or near the Solar System, where motions—within the precision of our measurements—remain Newtonian.

2. MOND

Well before Proxima Centauri—at around .111 ly (7300 AU), the strength of the Sun's gravitational field drops below the MOND constant a_0 of 1.2E−10 m s^{-2}. According to MOND, as discussed, this constant marks the transition to the non-Newtonian acceleration regime, but, as just demonstrated, the Sun's gravity remains Newtonian well past Proxima Centauri! In principle, MOND *does* account for such scenarios with a concept called the "external-field effect" (EFE)—where modified gravity is muted in systems that are in the presence of relatively strong external fields.

In contrast, under the present thesis, non-Newtonian effects in a galaxy are quantified by the frame constant k_f—which applies to all bodies and systems of bodies that are gravitationally bound to and part of the host galaxy. Looking at the Milky Way, at small scales—the scale of our Solar System and even out to nearby stars, k_f is too small for non-Newtonian effects to be observed. Non-Newtonian motion is only observed at galactic scales—where distances are such that the non-Newtonian acceleration component is dominant.

The vertical dashed cyan line in Fig. 1 is drawn at an H_f of 3.09 km s^{-1} Mpc^{-1}, which is the value of H_f that corresponds to the MOND constant a_0 of 1.2E−10 m s^{-2}. Equating the deep-MOND-regime relation, Eq. (10), with Eq. (10)—the expansion component of acceleration, distills as follows:

$$\begin{aligned} &a = \frac{\sqrt{a_0 m G}}{r} \ , \\ &a = \frac{2\sqrt{cH_M m G}}{r} \ , \\ &\frac{\sqrt{a_0 \cancel{m_t G}}}{\cancel{r}} = \frac{2\sqrt{cH_M \cancel{m_t G}}}{\cancel{r}} \ , \\ &a_0 = 4cH_M \ , \\ &H_M = \frac{1}{4c}a_0 \ , \\ &H_M = 3.09\,\mathrm{km\,s^{-1}Mpc^{-1}} \ . \end{aligned}$$

The expansion parameter corresponding to the MOND constant a_0 is designated as H_M. With Eq. (1), H_M equates to a frame speed v_M of 583 km s^{-1}, which is where the dashed blue line intersects the left red curve in Fig. 1. The blue line intersects the right magenta curve at 412 km s^{-1}. Interestingly, the Milky Way's peculiar speed of 565 km s^{-1} is near v_M, and correspondingly, its expansion parameter of 3.04 km s^{-1} Mpc^{-1} is close to H_M. The MOND constant a_0 was arrived at empirically and is the number that yields the best curve fits—on average—for the many galaxies tested against MOND. Ultimately, is a_0 simply a gauge of the average peculiar speed of galaxies in the cosmos?

II. THE GALACTIC MODEL

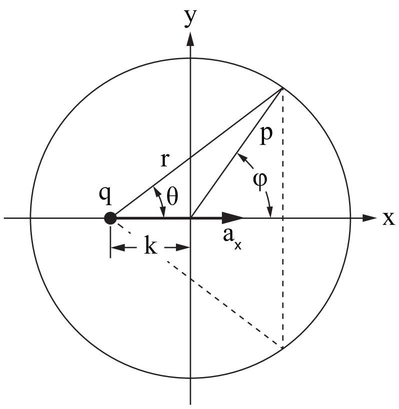


FIG. 2.

The galactic-rotation curves presented in Appendix III were generated according to the following method: The bulge is modeled as a sphere with a density that varies according to radius p, and the disk is modeled as a zero-thickness plate that varies in surface density according to radius p. Density, at a given radius, is constant with respect to angle, and thus, local features in the disk such as spiral arms and bars are ignored.

Both models are described by Fig. 2, where, for the bulge, the circle represents a layer of the sphere and the vertical dashed line represents the ring that is formed by the intersection of that layer at radius p and a plane perpendicular to axis x. For the disk, the circle marks a radial offset p in a top-down view of the disk. In both models, $\vec{r}$ is a vector between point q at offset k from the y axis and a mass–element at p.

For the bulge, the mass m_r of a ring at radius p is thus:

$$V_r = \frac{2\pi}{3}(\cos\phi_1 - \cos\phi_2)({p_2}^3 - {p_1}^3) , \tag{10}$$

$$m_r = \rho(p)V_r . \tag{11}$$

The volume V_r of the ring bound by radii p_1 and p_2, and angles phi_1 and phi_2 is given by Eq. (10). The mass is computed via Eq. (11), where the average density between p_1 and p_2 is multiplied by the volume. Notice that when $\phi_1 = 0$, $\phi_2 = \pi$, and $p_1 = 0$, V_r is the volume of a sphere of radius p_2.

Relative to point q at offset k, the distance r to a differential element of ring mass is constant with respect to the revolved angle about the x axis, and thus, the acceleration along $\vec{r}$ from the gravitational field of a ring element is equal in magnitude to that of all elements of equal size across this angle. With a vector pointing to each element, the sum of these vectors yields the net acceleration a_x at point q with respect to a ring of mass m_r. The radial components of the vector sum all cancel, leaving a_x pointing along the x axis as follows:

$$p = \frac{p_1 + p_2}{2} , \tag{12}$$

$$\phi = \frac{\phi_1 + \phi_2}{2} , \tag{13}$$

$$r = \sqrt{p^2 + 2p\cos\phi k + k^2} , \tag{14}$$

$$\cos\theta = \frac{p\cos\phi + k}{r} , \tag{15}$$

$$a_x = \frac{m_r G}{r}\left(\frac{1}{r} + \frac{2\sqrt{cH_f}}{\sqrt{m_t G}}\right)\cos\theta , \tag{16}$$

$$\frac{v^2}{r} = a_x , \tag{17}$$

$$v = \sqrt{a_x r} . \tag{18}$$

The acceleration at point q along $\vec{r}$ from a mass element of the ring is given by Eq. (2), and the sum of these acceleration vectors is given by Eq. (16). Notice that when point q is centered in the ring—where $\theta = \pi/2$—the net acceleration is zero, as expected. To compute only the Newtonian component of a_x, H_f, in Eq. (16), is set to zero. Eq. (17) is the familiar relation between tangential speed v—where the motion is circular—and centripetal acceleration, which, in this case, is a_x. The rotation curves were drawn with the assumption of circular motion—with the rotational velocity v given by Eq. (18).

To calculate the net acceleration a_x at a given offset k from the gravitational field of the full bulge, the bulge is divided into a series of thin layers—with each layer having a certain density $\rho(p)$. Each layer is divided into a series of thin rings orthogonal to the x axis. Eqs. (10) and (11) are used to calculate the mass of a ring defined by radii p_1 and p_2, and angles ϕ_1 and ϕ_2. Eqs. (12)–(16) are then used to calculate the acceleration a_x towards the ring. Running this calculation for small increments of ϕ from 0 to π at a given p_1 and p_2—which defines a shell—and summing a_x for each increment, gives the acceleration towards the shell. Running the shell calculation for small increments of p from 0 to the bulge radius, and summing the shell acceleration for each increment, gives the net acceleration a_b towards the bulge. As a test of the code, offset k can be placed inside and outside of the bulge and a_b can be tested to see if Newton's shell theorem is confirmed—with H_f in Eq. (16) set to zero.

The algorithm is nearly identical for the disk calculation. Instead of calculating a ring mass, however, the mass of a pair of arc elements—where one element is the mirror of the other about the x axis—is calculated as follows:

$$A_r = (\phi_2 - \phi_1)({p_2}^2 - {p_1}^2) , \tag{19}$$

$$m_r = \rho(p)A_r . \tag{20}$$

Looking down at the disk—the view depicted in Fig. 2, a finite area element A_r of the disk is defined by radii p_1 and p_2, and angles ϕ_1 and ϕ_2. The mass of this element m_r is given by Eq. (20), where this area is multiplied by the surface density $\rho(p)$ at radius p. Notice that in the relation for A_r—Eq. (19), the area element is double the size that would be contained within the arc between ϕ_1 and ϕ_2. The area A_r for the arc from 0 to π yields, with p_1 set to zero, the area of a circle of radius p_2 and not a half circle. A_r, and by extension, m_r, refer to an element–pair mirrored about the x axis.

Looking at Fig. 2, $\vec{r}$ points to a disk element, and the mirror of this vector, drawn as a dashed line, points to the reflected element in the pair. Because of the symmetry about the x axis, the radial components of the acceleration vectors from point q cancel, leaving a_x aligned along the x axis—as was the case for the bulge. Thus, m_r for a disk-element pair—as calculated by Eq. (20)—can be fed directly into Eqs. (12)–(16) to compute the gravitational acceleration a_x towards the pair. This calculation is run for small increments of ϕ from 0 to π at a given p_1 and p_2, which gives the acceleration towards a ring of the disk. Integrating across the disk from zero to the disk radius by small increments of p and summing a_x for each ring gives the net acceleration a_d towards the disk.

III. GALACTIC ROTATION CURVES

TABLE I. A Selection of Nearby SPARC Galaxies (mass models).

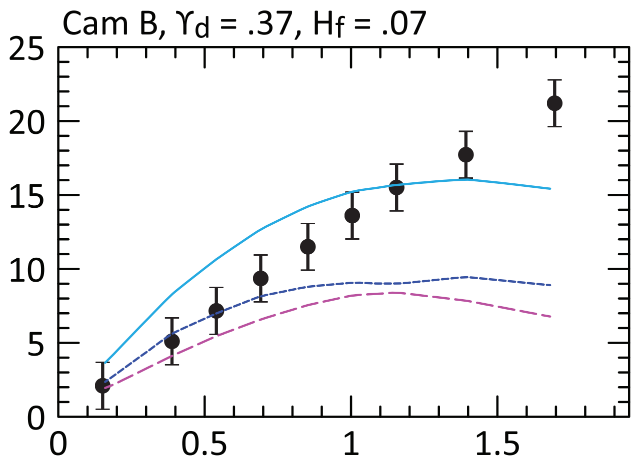
Galaxy	Dark-matter model	Υ_b	Υ_d	Distance (Mpc)	CMB redshift (km s^{-1})	r_{max} (kpc)
Cam B	NFW flat	0	.37	3.5	23	1.68
DDO 064	Burkert	0	.49	3.800-7.110	784	3.03
DDO 154	coreNFW flat	0	.29	4.04	639	5.97
DDO 168	Lucky13 LCDM	0	.55	4.25	380	4.07
ESO 444-G084	NFW LCDM	0	.54	4.53	870	4.42
IC 2574	DC14 flat	0	.44	3.890-3.940	148	10.20
NGC 0055	Einasto flat	0	.49	1.850-2.340	115	13.36
NGC 0247	coreNFW LCDM[a]	0	1.0	3.270-3.670	143	14.50
NGC 0300	Lucky13 LCDM	0	.73	1.790-2.170	91	12.04
NGC 2366	Burkert	0	.38	3.200-3.340	134	6.02
NGC 2403	Burkert	0	.90	3.010-3.930	182	19.60
NGC 2915	Burkert	0	.42	3.580-4.290	588	9.90
NGC 2976	Burkert	0	.48	3.520-3.630	90	2.25
NGC 3109	NFW LCDM	0	.51	1.333	738	7.16
NGC 3741	Burkert	0	.77	3.150-3.230	456	7.13
NGC 4068	coreNFW LCDM	0	.45	4.3	784	2.28
NGC 4214	NFW LCDM	0	.54	2.700-2.930	550	5.63
NGC 6789	DC14 LCDM[a]	0	1.0	3.6	275	0.73
NGC 6946	Burkert	.63	.58	4.51	133	18.90
NGC 7793	Burkert	0	.56	3.390-3.840	53	7.97
UGC 04305	Burkert	0	.50	3.24	203	5.45
UGC 04483	Burkert	0	.50	3.2	213	1.21
UGC 07232	Burkert	0	.50	2.95	486	.82
UGC 07524	DC14 flat	0	.43	4.410-4.610	585	10.50
UGC 07559	coreNFW LCDM	0	.43	4.97	470	2.50
UGC 07577	coreNFW LCDM	0	.37	2.6	417	1.50
UGC 07866	coreNFW LCDM	0	.48	4.57	592	2.29
UGC 08490	DC14 LCDM	0	.96	4.790-5.550	322	12.50
UGCA 444	Burkert	0	.49	0.93	457	2.60

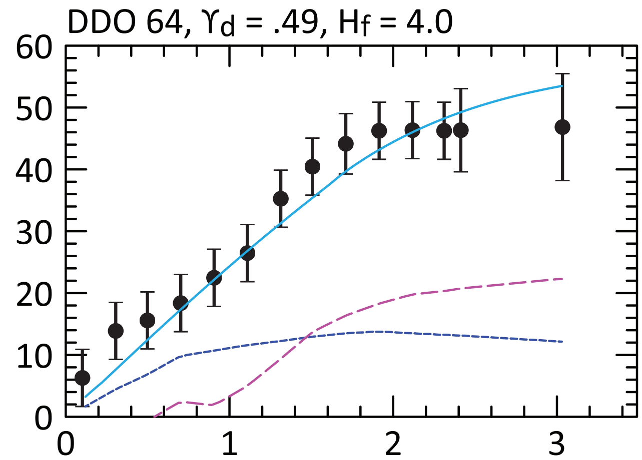
[a] Error bars are from this dark-matter model, but the mass-to-light ratio is less than that given by the model.
[b] Mass-to-light ratios were sourced from the SPARC database [1].
[c] Distance and redshift data were sourced from the *NASA/IPAC Extragalactic Database* [2].

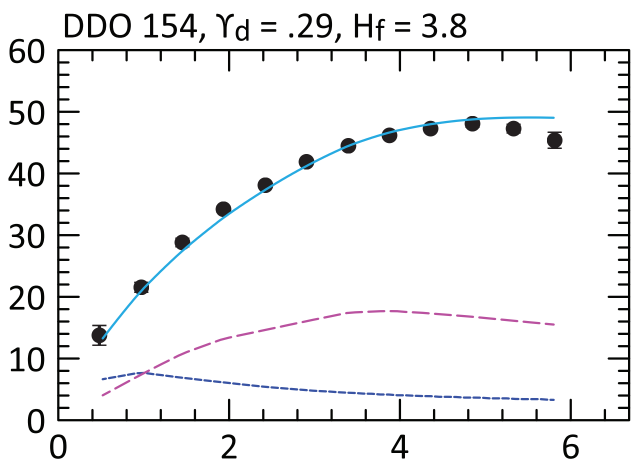
TABLE II. A Selection of Nearby SPARC Galaxies (H_f and k_f).

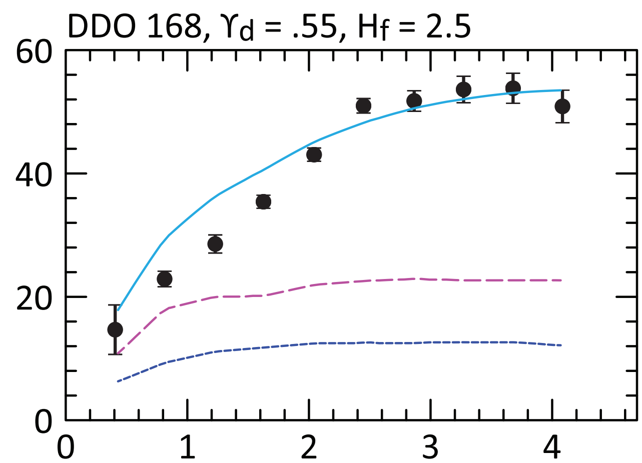
Galaxy	Mass (kg)	H_f (km s^{-1} Mpc^{-1})	k_f (kg^{-1} s^{-1})	v_{flat} (km s^{-1})	Poor fit	Outlier
Cam B	7.12E37	.07	3.19E−59	11	x	
DDO 064	6.90E38	4.0	1.88E−58	52		
DDO 154	4.75E38	3.8	2.59E−58	47		
DDO 168	7.69E38	2.5	1.05E−58	47		
ESO 444-G084	4.49E38	4.0	2.89E−58	46		
IC 2574	3.24E39	1.8	1.80E−59	62		
NGC 0055	8.97E39	1.3	4.70E−60	74		
NGC 0247	1.62E40	1.5	3.00E−60	89		
NGC 0300	6.21E39	.90	4.70E−60	62		
NGC 2366	1.48E39	1.3	2.85E−59	47		
NGC 2403	2.52E40	3.1	3.99E−60	119		x
NGC 2915	1.56E39	12	2.49E−58	83	x	x
NGC 2976	2.30E39	1.9	2.68E−59	58		
NGC 3109	1.35E39	3.6	8.64E−59	60		
NGC 3741	4.99E38	5.0	3.25E−58	50		
NGC 4068	4.84E38	2.5	1.67E−58	42		
NGC 4214	1.44E39	3.0	6.75E−59	58		
NGC 6789	2.05E38	3.0	4.74E−58	36		
NGC 6946	7.23E40	1.1	4.93E−61	120		
NGC 7793	9.83E39	.35	1.15E−60	55		
UGC 04305	1.70E39	2.0	3.81E−59	54	x	
UGC 04483	8.18E37	1.8	7.13E−58	25		
UGC 07232	1.74E38	4.0	7.45E−58	37		x
UGC 07524	6.06E39	3.0	1.60E−59	83	x	
UGC 07559	3.10E38	1.5	1.57E−58	33		x
UGC 07577	7.55E37	.70	3.00E−58	19		x
UGC 07866	2.23E38	3.0	4.36E−58	36		
UGC 08490	4.95E39	2.2	1.44E−59	73		
UGCA 444	1.88E38	3.2	5.52E−58	35		

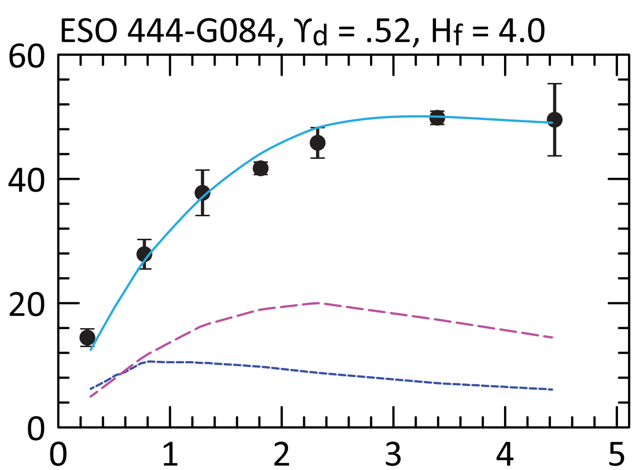
In the following graphs, the units for the vertical axes are km s^{-1}, and the units for the horizontal axes are kpc. The Newtonian curve for the stellar disk is drawn with a dashed blue line. The Newtonian contribution from the gas disk is drawn with a dashed magenta line with larger dashes, and the bulge contribution is drawn as a dotted green line. The net rotation curve, generated via Eq. (2), is drawn as a solid cyan line.

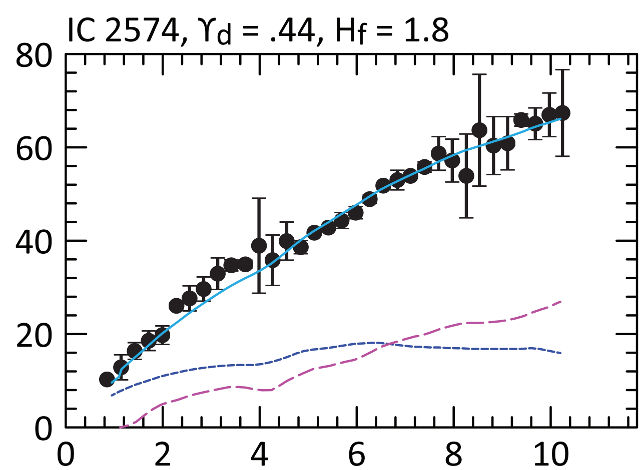


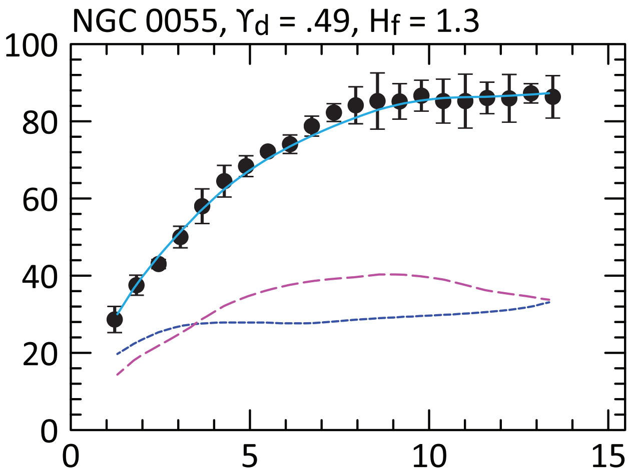








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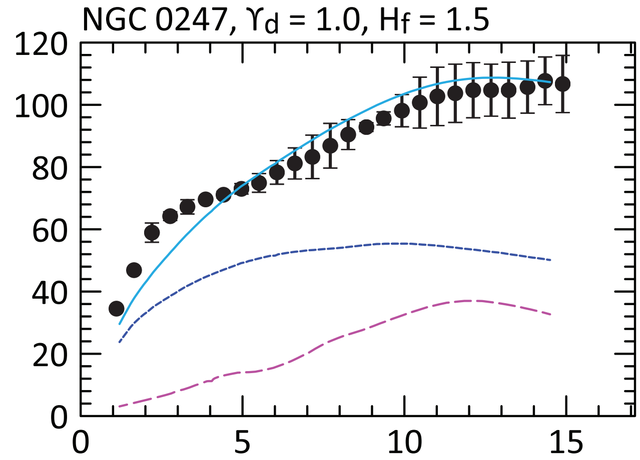
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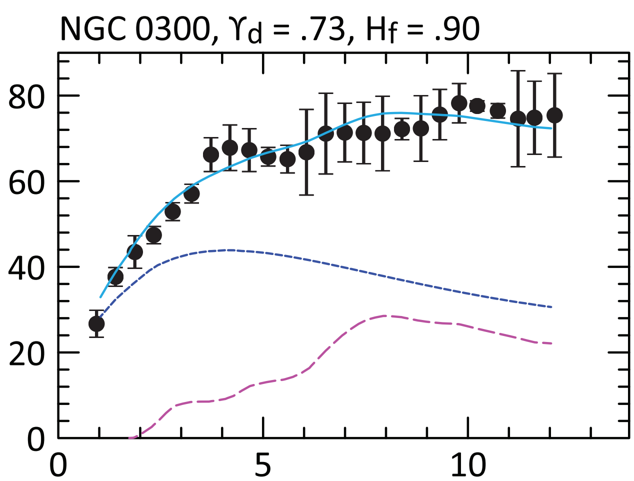
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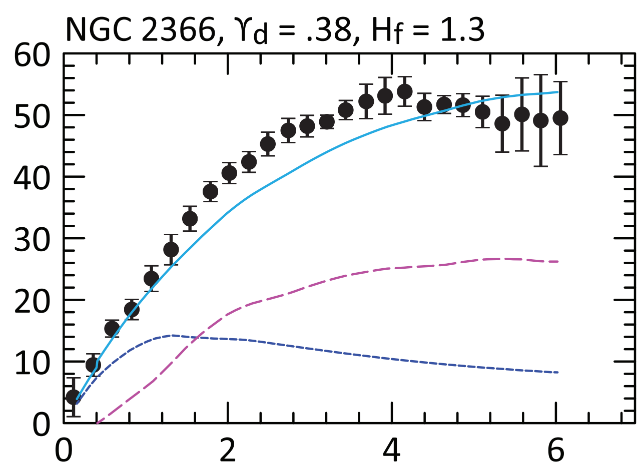
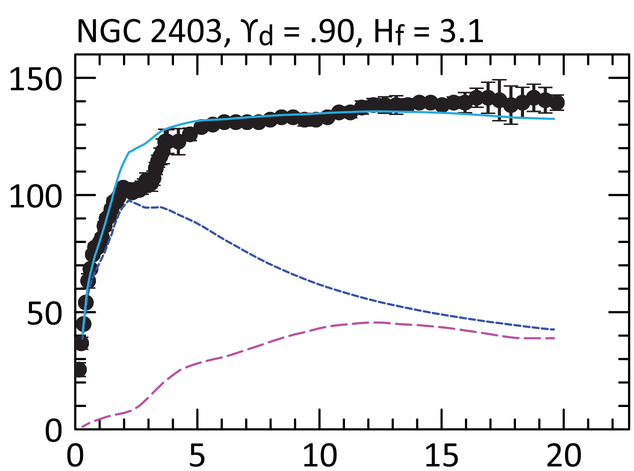
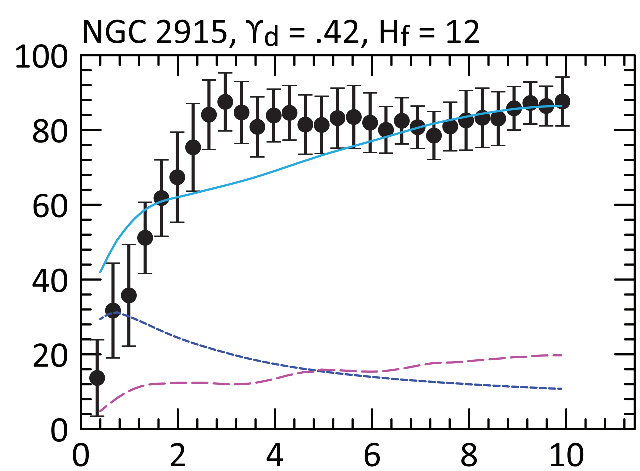
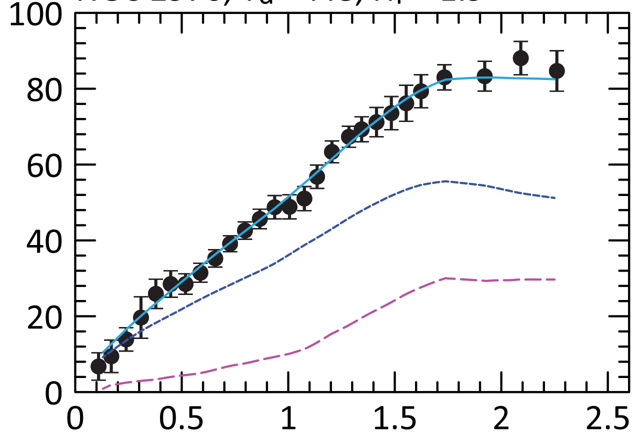

NGC 2366, ϒd = .38, Hf = 1.3

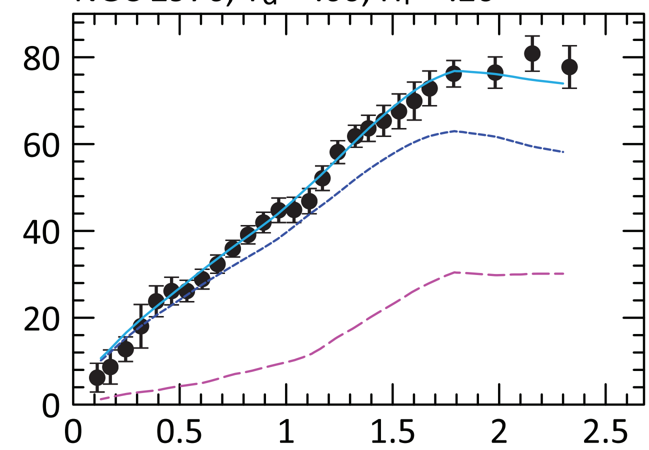

NGC 2403, ϒd = .90, Hf = 3.1


NGC 2915, ϒd = .42, Hf = 12

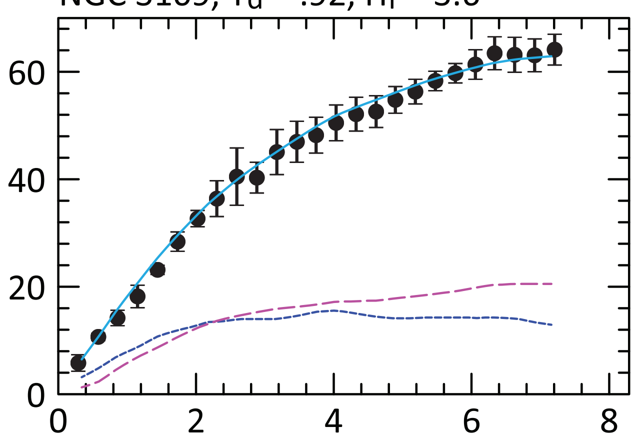
NGC 2976, Υ_d = .48, H_f = 1.9



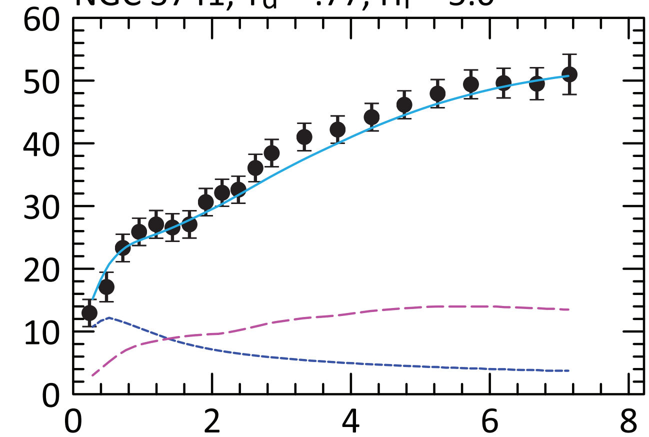
NGC 2976, Υ_d = .60, H_f = .20



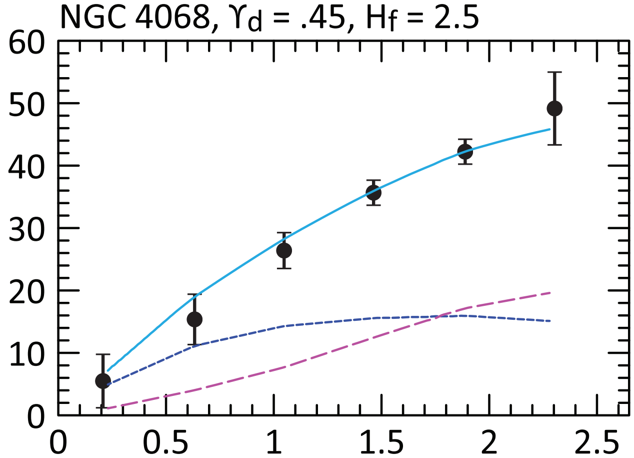
NGC 3109, Υ_d = .92, H_f = 3.6



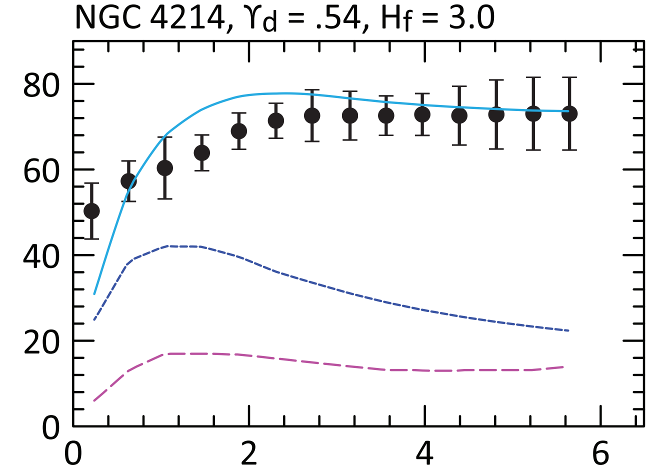
NGC 3741, Υ_d = .77, H_f = 5.0

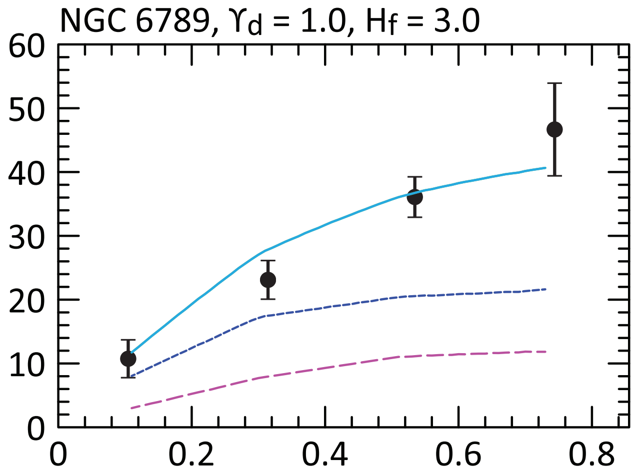


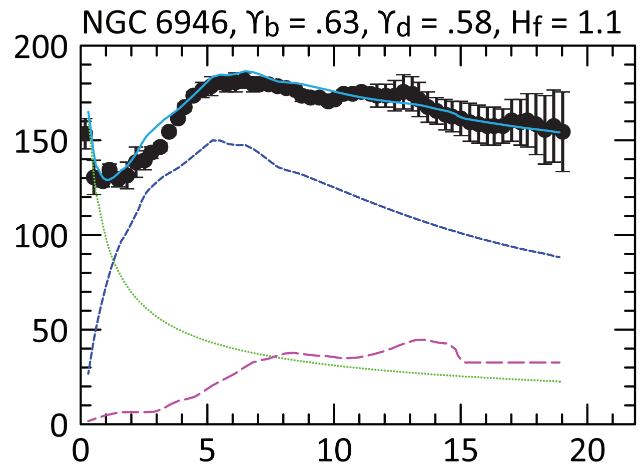
NGC 4068, Υ_d = .45, H_f = 2.5

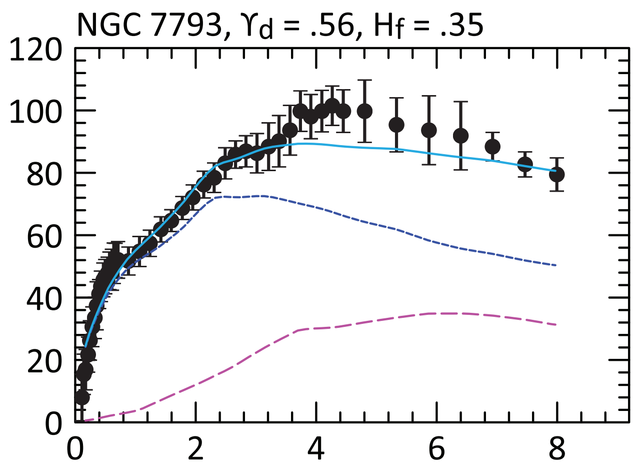


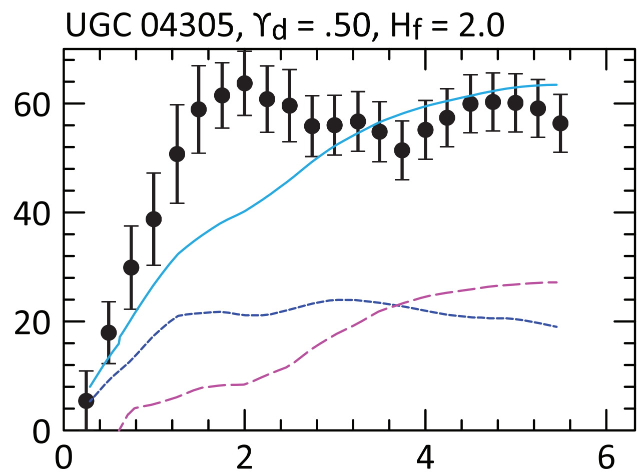
NGC 4214, Υ_d = .54, H_f = 3.0

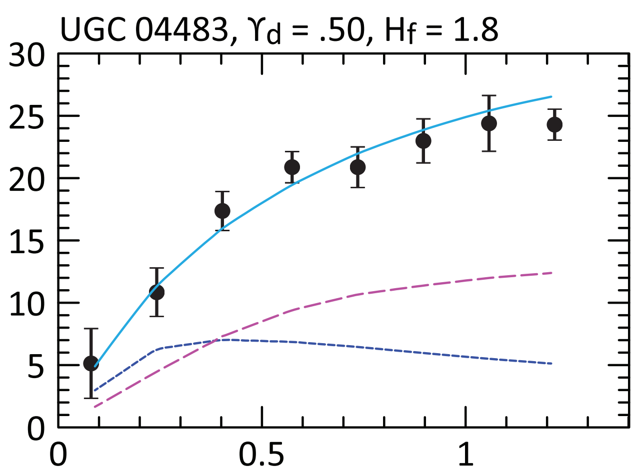


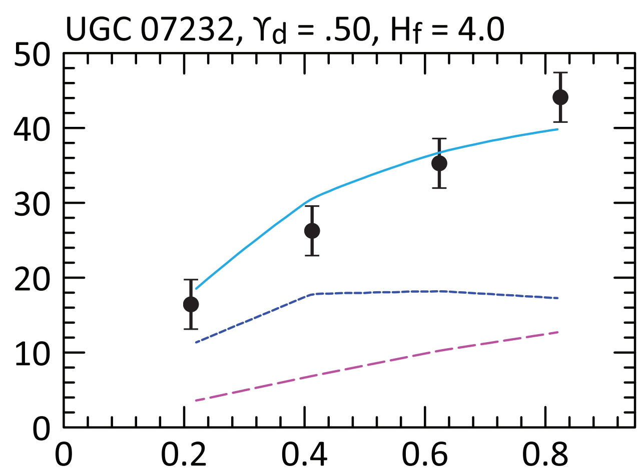

NGC 6789, Υ_d = 1.0, H_f = 3.0
0
10
20
30
40
50
60
0
0.2
0.4
0.6
0.8


NGC 6946, Υ_b = .63, Υ_d = .58, H_f = 1.1
0
50
100
150
200
0
5
10
15
20


NGC 7793, Υ_d = .56, H_f = .35
0
20
40
60
80
100
120
0
2
4
6
8


UGC 04305, Υ_d = .50, H_f = 2.0
0
20
40
60
0
2
4
6


UGC 04483, Υ_d = .50, H_f = 1.8
0
5
10
15
20
25
30
0
0.5
1


UGC 07232, Υ_d = .50, H_f = 4.0
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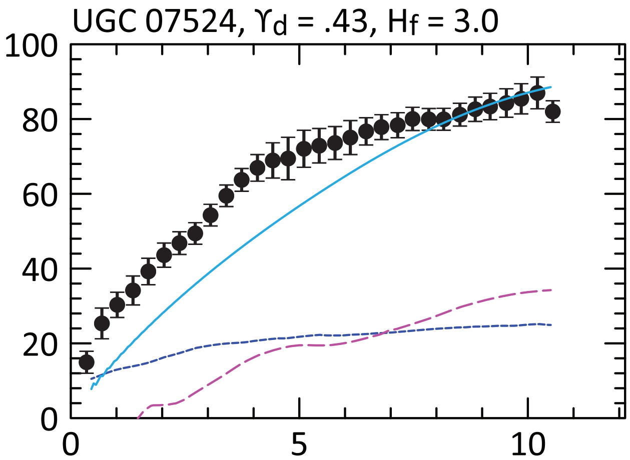
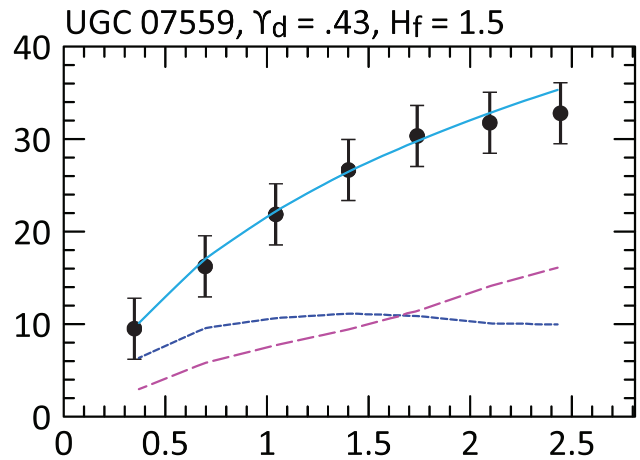
10
20
30
40
50
0
0.2
0.4
0.6
0.8


UGC 07524, Υ_d = .43, H_f = 3.0
100
80
60
40
20
0
0
5
10


UGC 07559, Υ_d = .43, H_f = 1.5
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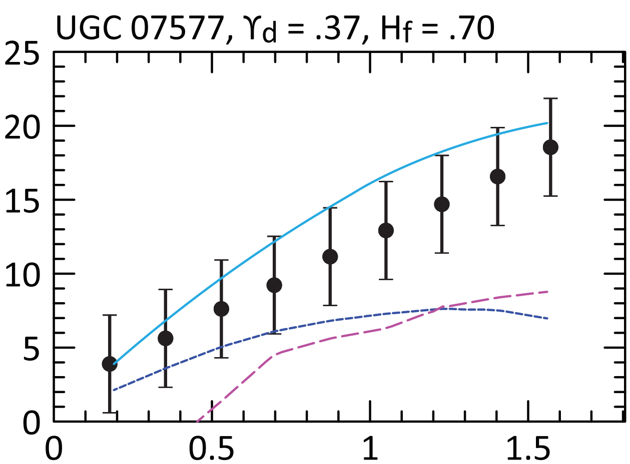
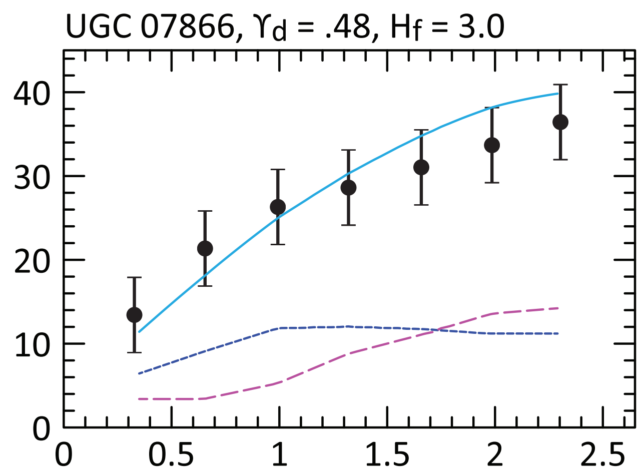
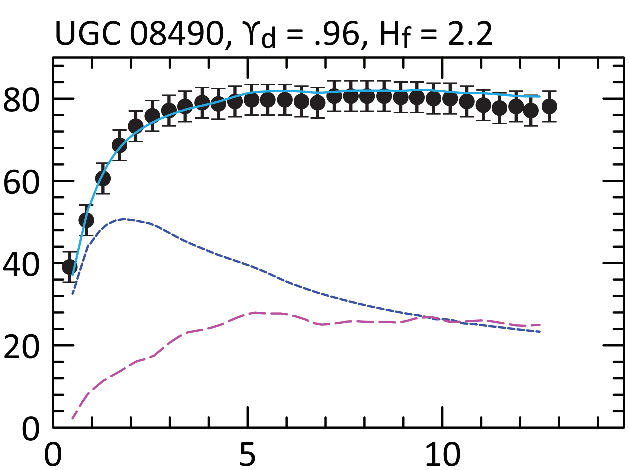
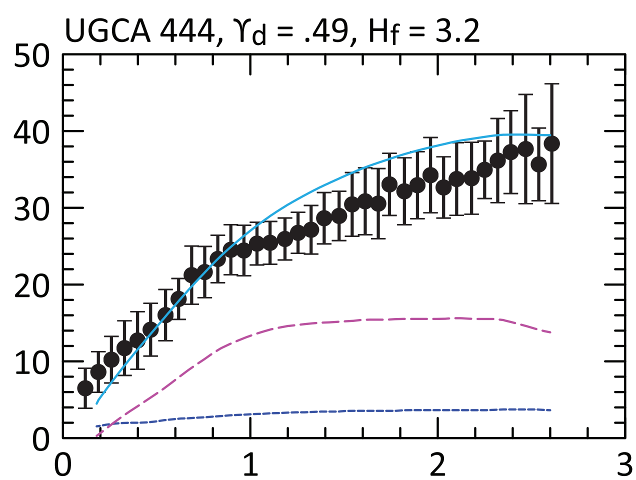
30
20
10
0
0
0.5
1
1.5
2
2.5


UGC 07577, Υ_d = .37, H_f = .70
25
20
15
10
5
0
0
0.5
1
1.5


UGC 07866, Υ_d = .48, H_f = 3.0
40
30
20
10
0
0
0.5
1
1.5
2
2.5


UGC 08490, Υ_d = .96, H_f = 2.2
80
60
40
20
0
0
5
10


UGCA 444, Υ_d = .49, H_f = 3.2
50
40
30
20
10
0
0
1
2
3

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